

Lines, Triangles & Polygons



Parallel lines and transversals

- 1 Two parallel lines are cut by a transversal, forming an angle of 105° on one line. Find the corresponding angle, the alternate interior angle, and the co-interior (same-side interior) angle on the other line.
 - 2 Two angles formed by a transversal crossing parallel lines are alternate exterior angles, so they are equal. If one measures $3x + 10$ and the other measures $5x - 30$, find x and the measure of the angle.
 - 3 Two parallel lines are cut by a transversal. One interior angle is $4x + 15$, and its corresponding angle on the other line is $6x - 25$. Find x and the angle measure.
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Triangle angle sum and the exterior angle theorem

- 4 The angles of a triangle are in the ratio $3 : 5 : 7$. Find each angle.
 - 5 In triangle ABC , $\angle A = 52^\circ$ and the exterior angle at C is 130° . Find $\angle B$.
 - 6 An exterior angle of a triangle measures 115° , and one remote interior angle is 50° . Find the other remote interior angle and the third (adjacent) interior angle.
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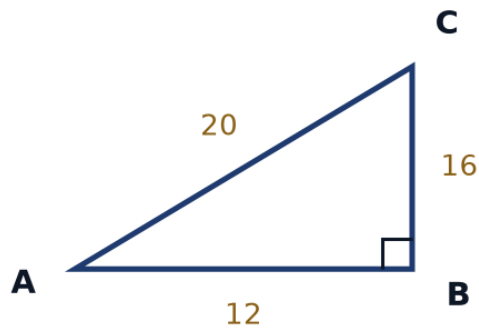
Triangle congruence criteria

- 7 State which congruence criterion (SSS, SAS, ASA, AAS, or RHS) proves triangles PQR and XYZ congruent, given $PQ = XY$, $\angle Q = \angle Y$, and $QR = YZ$.
 - 8 Triangle DEF has $DE = 9$, $EF = 12$, $\angle E = 90^\circ$, and triangle GHI has $GH = 9$, $HI = 12$, $\angle H = 90^\circ$. Are the triangles congruent? By which criterion?
 - 9 Two triangles have all three pairs of corresponding sides equal in length. Which congruence criterion applies, and are the triangles necessarily congruent?
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Similar triangles and proportional reasoning

- 10 Triangle $ABC \sim$ triangle DEF , with $AB = 6$, $BC = 8$, and $DE = 9$. Find EF .

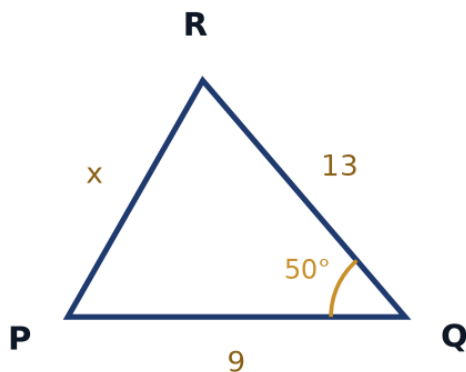
- 11 Triangle ABC shown below has $AB = 12$, $BC = 16$, and $CA = 20$. It is similar to triangle DEF , where $DE = 15$ corresponds to AB . Find EF and FD .



- 12 Triangle JKL has $JK = 15$ and $KL = 21$, and is similar to triangle MNO with scale factor $\frac{2}{3}$ (triangle MNO is smaller), where MN corresponds to JK . Given $MN = 10$, verify the scale factor is consistent, then find NO .

The Law of Cosines

- 13 In triangle PQR shown below, $PQ = 9$, $QR = 13$, and $\angle Q = 50^\circ$. Find $PR = x$, to the nearest hundredth.



- 14 A triangle has sides $a = 7$, $b = 8$, $c = 10$. Use the Law of Cosines to find angle C (opposite side c), to the nearest tenth of a degree.
- 15 A triangle has sides $a = 6$, $b = 6$, and included angle $C = 80^\circ$. Find side c , to the nearest hundredth, and state what type of triangle this is based on the two given equal sides.

Heron's formula

- 16 Find the area of a triangle with sides 13, 14, 15 using Heron's formula.
- 17 A triangular plot of land has sides 9, 10, 17. Find its area using Heron's formula.

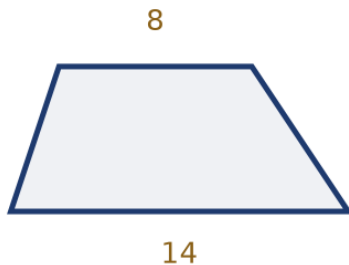
- 18 A triangular sail has sides 5 m, 12 m, and 13 m. Find its area using Heron's formula, and verify using the right-triangle area formula (since 5-12-13 is a Pythagorean triple).
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Interior and exterior angles of polygons

- 19 Find the sum of the interior angles of a regular decagon (10 sides), and the measure of each interior angle.
- 20 Each exterior angle of a regular polygon measures 24° . How many sides does the polygon have?
- 21 A regular polygon has an interior angle of 150° . How many sides does it have?
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Area of polygons

- 22 The trapezoid below has parallel sides of length 14 and 8, and height 6. Find its area.



- 23 Find the area of a regular hexagon with side length 6.
- 24 A composite figure is an 8-by-10 rectangle with a 3-by-4 rectangular notch removed from one corner. Find the remaining area.
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A well-designed word problem: a surveyor's triangular lots

- 25 This question has three parts. A surveyor measures a triangular lot with sides 30 m, 40 m, and 50 m. (a) Verify using the Pythagorean theorem that the lot is a right triangle. (b) Find the area of the lot using the right-triangle area formula, then confirm your answer using Heron's formula. (c) A second, similar triangular lot has its shortest side measuring 45 m. Using the similarity ratio, find the area of the second lot.
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Multiple choice — easy

- 26 The angles of a triangle are in the ratio $2 : 1 : 3$. Find each angle.
- a. 30, 64, 86
 - b. 30, 60, 90
 - c. 32, 60, 88
 - d. 90, 60, 30
- 27 The angles of a triangle are in the ratio $5 : 9 : 6$. Find each angle.
- a. 45, 58, 77
 - b. 81, 54, 45
 - c. 45, 54, 81
 - d. 47, 54, 79
- 28 The angles of a triangle are in the ratio $9 : 3 : 3$. Find each angle.
- a. 38, 36, 106
 - b. 36, 40, 104
 - c. 36, 36, 108
 - d. 108, 36, 36
- 29 The angles of a triangle are in the ratio $6 : 5 : 7$. Find each angle.
- a. 50, 64, 66
 - b. 52, 60, 68
 - c. 50, 60, 70
 - d. 70, 60, 50

- 30** The angles of a triangle are in the ratio 1 : 5 : 6. Find each angle.
- a. 90, 75, 15
 - b. 15, 79, 86
 - c. 15, 75, 90
 - d. 17, 75, 88
- 31** The angles of a triangle are in the ratio 2 : 3 : 5. Find each angle.
- a. 38, 54, 88
 - b. 36, 58, 86
 - c. 36, 54, 90
 - d. 90, 54, 36
- 32** The angles of a triangle are in the ratio 4 : 5 : 6. Find each angle.
- a. 50, 60, 70
 - b. 48, 64, 68
 - c. 48, 60, 72
 - d. 72, 60, 48
- 33** The angles of a triangle are in the ratio 5 : 8 : 2. Find each angle.
- a. 24, 60, 96
 - b. 24, 64, 92
 - c. 26, 60, 94
 - d. 96, 60, 24

- 34** The angles of a triangle are in the ratio $9 : 2 : 9$. Find each angle.
- a. 81, 81, 18
 - b. 18, 81, 81
 - c. 20, 81, 79
 - d. 18, 85, 77
- 35** The angles of a triangle are in the ratio $3 : 3 : 9$. Find each angle.
- a. 108, 36, 36
 - b. 38, 36, 106
 - c. 36, 40, 104
 - d. 36, 36, 108
- 36** The angles of a triangle are in the ratio $2 : 3 : 1$. Find each angle.
- a. 30, 64, 86
 - b. 90, 60, 30
 - c. 32, 60, 88
 - d. 30, 60, 90
- 37** The angles of a triangle are in the ratio $9 : 5 : 6$. Find each angle.
- a. 45, 54, 81
 - b. 81, 54, 45
 - c. 47, 54, 79
 - d. 45, 58, 77

38 The angles of a triangle are in the ratio 1 : 2 : 6. Find each angle.

- a. 22, 40, 118
- b. 20, 40, 120
- c. 120, 40, 20
- d. 20, 44, 116

39 The angles of a triangle are in the ratio 5 : 6 : 1. Find each angle.

- a. 15, 75, 90
- b. 17, 75, 88
- c. 90, 75, 15
- d. 15, 79, 86

40 The angles of a triangle are in the ratio 8 : 3 : 4. Find each angle.

- a. 96, 48, 36
- b. 38, 48, 94
- c. 36, 48, 96
- d. 36, 52, 92

Multiple choice — medium

41 An exterior angle of a triangle measures 82° , and one remote interior angle is 30° . Find the other remote interior angle.

- a. 52°
- b. 57°
- c. 47°
- d. 30°

- 42 An exterior angle of a triangle measures 90° , and one remote interior angle is 39° . Find the other remote interior angle.
- a. 56°
 - b. 46°
 - c. 51°
 - d. 39°
- 43 An exterior angle of a triangle measures 67° , and one remote interior angle is 25° . Find the other remote interior angle.
- a. 42°
 - b. 25°
 - c. 37°
 - d. 47°
- 44 An exterior angle of a triangle measures 76° , and one remote interior angle is 44° . Find the other remote interior angle.
- a. 27°
 - b. 44°
 - c. 37°
 - d. 32°
- 45 An exterior angle of a triangle measures 116° , and one remote interior angle is 67° . Find the other remote interior angle.
- a. 54°
 - b. 49°
 - c. 44°
 - d. 67°

- 46 An exterior angle of a triangle measures 56° , and one remote interior angle is 28° . Find the other remote interior angle.
- a. 28°
 - b. 31°
 - c. 23°
 - d. 33°
- 47 An exterior angle of a triangle measures 92° , and one remote interior angle is 48° . Find the other remote interior angle.
- a. 49°
 - b. 48°
 - c. 44°
 - d. 39°
- 48 An exterior angle of a triangle measures 54° , and one remote interior angle is 23° . Find the other remote interior angle.
- a. 23°
 - b. 36°
 - c. 31°
 - d. 26°
- 49 An exterior angle of a triangle measures 94° , and one remote interior angle is 67° . Find the other remote interior angle.
- a. 67°
 - b. 27°
 - c. 32°
 - d. 22°

- 50 An exterior angle of a triangle measures 95° , and one remote interior angle is 26° . Find the other remote interior angle.
- a. 26°
 - b. 69°
 - c. 64°
 - d. 74°
- 51 An exterior angle of a triangle measures 88° , and one remote interior angle is 22° . Find the other remote interior angle.
- a. 22°
 - b. 71°
 - c. 61°
 - d. 66°
- 52 An exterior angle of a triangle measures 51° , and one remote interior angle is 31° . Find the other remote interior angle.
- a. 31°
 - b. 20°
 - c. 25°
 - d. 15°
- 53 An exterior angle of a triangle measures 86° , and one remote interior angle is 54° . Find the other remote interior angle.
- a. 32°
 - b. 54°
 - c. 27°
 - d. 37°

- 54 An exterior angle of a triangle measures 89° , and one remote interior angle is 21° . Find the other remote interior angle.
- a. 21°
 - b. 68°
 - c. 73°
 - d. 63°
- 55 An exterior angle of a triangle measures 47° , and one remote interior angle is 25° . Find the other remote interior angle.
- a. 25°
 - b. 22°
 - c. 27°
 - d. 17°
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Multiple choice — hard

- 56 A triangle has sides $a = 6$, $b = 11$, and included angle $C = 110^\circ$. Find side c , to the nearest hundredth.
- a. 13.72
 - b. 14.72
 - c. 14.22
 - d. 15.22
- 57 A triangle has sides $a = 12$, $b = 10$, and included angle $C = 40^\circ$. Find side c , to the nearest hundredth.
- a. 8.26
 - b. 8.76
 - c. 7.26
 - d. 7.76

- 58 A triangle has sides $a = 7$, $b = 14$, and included angle $C = 110^\circ$. Find side c , to the nearest hundredth.
- a. 18.16
 - b. 17.16
 - c. 18.66
 - d. 17.66
- 59 A triangle has sides $a = 7$, $b = 13$, and included angle $C = 80^\circ$. Find side c , to the nearest hundredth.
- a. 13.65
 - b. 14.15
 - c. 14.65
 - d. 13.15
- 60 A triangle has sides $a = 10$, $b = 6$, and included angle $C = 70^\circ$. Find side c , to the nearest hundredth.
- a. 10.74
 - b. 9.74
 - c. 9.24
 - d. 10.24
- 61 A triangular plot of land has sides 12, 35, 37. Find its area using Heron's formula.
- a. 42
 - b. 210
 - c. 212
 - d. 208

- 62 A triangular plot of land has sides 24, 45, 51. Find its area using Heron's formula.
- a. 542
 - b. 60
 - c. 540
 - d. 538
- 63 A triangular plot of land has sides 9, 12, 15. Find its area using Heron's formula.
- a. 52
 - b. 56
 - c. 54
 - d. 18
- 64 A triangular plot of land has sides 36, 105, 111. Find its area using Heron's formula.
- a. 1890
 - b. 1888
 - c. 126
 - d. 1892
- 65 A triangular plot of land has sides 21, 72, 75. Find its area using Heron's formula.
- a. 758
 - b. 754
 - c. 84
 - d. 756

- 66** Find the sum of the interior angles of a regular polygon with 20 sides, and the measure of each interior angle.
- a. 3060, 162
 - b. 3240, 162
 - c. 3420, 157
 - d. 3240, 167
- 67** Find the sum of the interior angles of a regular polygon with 72 sides, and the measure of each interior angle.
- a. 12780, 170
 - b. 12600, 175
 - c. 12600, 180
 - d. 12420, 175
- 68** Find the sum of the interior angles of a regular polygon with 4 sides, and the measure of each interior angle.
- a. 360, 95
 - b. 180, 90
 - c. 360, 90
 - d. 540, 85
- 69** Find the sum of the interior angles of a regular polygon with 30 sides, and the measure of each interior angle.
- a. 5220, 163
 - b. 5040, 168
 - c. 4860, 168
 - d. 5040, 173

- 70** Find the sum of the interior angles of a regular polygon with 6 sides, and the measure of each interior angle.
- a.** 540, 120
 - b.** 900, 115
 - c.** 720, 120
 - d.** 720, 125